

Status — one line per numbered result

The short answer. PaperInventory.md is the long one and stays as the audit trail; this file is the thing to read first.

Measured on main after merging r0049's agents 3–7: build **1372 jobs, 0 errors, 0 warnings**, sorry **0**, axiom **0**, sorryAx **0**. Package: 124 modules, 46,049 lines, 2,119 theorems.

r0049 is not finished. agent8's stream — the necessity-site adjudication, the scoped-claims convention, and the standing staleness check — is still running and is not merged here. Every count below is main as of that partial merge.

Labels

Every paper property carries one of five labels. They answer two questions — **is it stated in Lean, and is it proved** — and the pair matters because **sorry can only see the second**.

#	Label	Stated?	Proved?	Meaning
1	S+P	yes, as the paper states it	yes	done
2	S+H	yes, as the paper states it	no — the proof is open	a real Lean statement with a hole. H is for <i>hypothesis</i> : the result is available only conditionally, as a theorem taking the open claim as an argument
3	S≠	yes, but not the paper's statement	yes	a weakening, an added hypothesis, or a deliberate repair of a printed defect
4	P	no — prose only	—	asserted in a docstring, never under the kernel
5	N	no statement at all	—	nothing in Lean mentions it

Two consequences worth stating once:

- **sorry 0 does not mean “everything is proved.”** A sorry is a hole in a proof someone *started*. Rows 4 and 5 were never started, and row 2's holes are carried as explicit hypotheses rather than as sorry, so none of the three appears in a sorry count. That is why this file reports rows 8–11 of the counts separately.
- **S≠ is not automatically a defect.** Six of the 14 are repairs of the paper's own printed errors, which is correct work; 8 are ours. r0049 re-derived the row population against the built environment rather than inheriting r0044's table, and it moved on both counts: four rows left the census (two closed in r0049, two closed in r0047 and never re-counted), and Theorem 26 moved from *ours* to *repair* once the printed statement was refuted.

Counts

#		Count
1	Numbered results in Gunter & Scott	30, verified — 16 Theorems, 13 Lemmas, 1 Proposition, 0 Corollaries; numbers 1–30 contiguous, no gap, no repeat
2	— fully proved	25
3	— partly proved, some part open	2 (results 7, 29)
4	— refuted as stated (our transcription or the paper)	3 (results 26, 28, 30)
5	— all 30 are now stated in the tree	0 missing
6	Paper properties (numbered conjuncts + prose claims)	239
7	— S+P stated and proved	169
8	— S+H stated, proof open	12 — <i>not re-derived after r0049; carried from r0047</i>
9	— S≠ stated, but not as the paper states it	14 (8 ours, 6 repairs)
10	— P + N no Lean statement at all	36
11	Prop-valued claims nothing proves	7 (8 strict)

Rows 8 and 11 are the two kinds of unfinished work, and they are different: row 8 is a real Lean statement whose proof is open; row 11 is a def naming a claim **nobody attempted**. **sorry 0 sees neither** — which is why both are counted here and neither shows up in a build.

The 30 numbered results

P proved · p partly proved · R refuted as stated · – not in the tree

#	Result	Status
1	Theorem 1	P least fixed point, and below every fixed point
2	Theorem 2 (Schröder-Bernstein)	P “Let S and T be sets. If $f : S \rightarrow T$ and $g : T \rightarrow S$ are injections, then there is a bijection $h : S \rightarrow T$ ” — printed folio 6. Genuinely absent until r0048 ; now R48.Agent1.theorem2. The inventory had mapped it to Mathlib’s <code>Function.Embedding.schröder_bernstein</code> , but that is Zermelo’s proof from Knaster–Tarski , where Gunter & Scott derive it from Theorem 1 — and <code>FixedPoint.lean</code> records that neither implies the other. The new proof follows the paper step for step, via the operator $Y \mapsto (T - f^*(S)) \cup f^*(g^*(Y))$
3	Theorem 3	P theorem3, theorem3_existsUnique
4	Lemma 4	P NormalSubposet.lean
5	Lemma 5	P FinitaryProjection.lean
6	Theorem 6	P theorem6
7	Theorem 7	p sentence 1 proved; sentences 2–3 proved only at the degenerate strength — every domain has an EffectivePresentation because <code>Classical.dec</code> fills its fields. The recursive forms are open: <code>Theorem7ArrowRecursive</code> , <code>Theorem7StrictRecursive</code> . r0049 restated <code>StepFunctionsDecidable</code> over <code>IsStepEnumeration</code> , quantified existentially, so the claim no longer names a guard its own hypotheses do not determine — R49.Agent3.stepFunctionsDecidable_of_compactGuard proves $\text{old} \rightarrow \text{new}$, a weakening . r0049 also supplied the recursion theory the residue needed: the <code>Finset (ℕ × ℕ)</code> coding is <code>Primrec</code> , the normal-subposet search is total, and §3.2’s two conditions decide boundedness and compute the join’s index (R49.Agent4.computablePred_bddAbove, <code>computable_joinIdx</code>). One item remains: <code>RecursiveNormal</code> for $K(D \rightarrow E)$
8	Lemma 8	P Product.lean
9	Lemma 9	P lem9_* — two printed misprints repaired (items 3 and 5)
10	Lemma 10	P 7 conjuncts, lem10_{prod, smash, sum, separated, lift, strict}
11	Theorem 11	P thm11, with thm11_converse
12	Theorem 12	P thm12 and the three powerdomains
13	Lemma 13	P lem13_smyth, lem13_hoare
14	Theorem 14	P thm14, both directions
15	Proposition 15	P “A bounded complete domain is bifinite” — printed folio 31. prop15 at <code>Skeleton/Section6.lean</code> :134, proved since that file was written. It is a Proposition, not a Theorem or Lemma , which is the whole reason it read as missing: both instruments that produced the gap searched only for <code>thm theorem lem lemma</code>
16	Theorem 16	P thm16, thm16_positive
17	Lemma 17	P 10 conjuncts. r0047 removed [BoundedComplete β] from <code>lem17_fun</code> and <code>lem17_strictFun</code>
18	Theorem 18	P thm18 — closed r0042, [propext, Classical.choice, Quot.sound]
19	Lemma 19	P lem19
20	Lemma 20	P lem20
21	Theorem 21	P thm21
22	Theorem 22	P thm22
23	Lemma 23	P lem23
24	Lemma 24	P lem24 (Gunter & Scott’s; not Gunter 1987’s Lemma 24, below)
25	Theorem 25	P thm25, thm25_isUniversal

#	Result	Status
26	Theorem 26	R refuted as printed (R49.Agent7.not_thm26Printed_of_two_zero_aritys), and proved as repaired — thm26, whose added $hs : \forall i, 0 < s_i$ is therefore a repair of a printed defect, not a defect of ours . Theorem 26 as printed is false for any signature with two or more 0-ary slots , including the paper’s own worked $(2, 0, 0, 0, 0, 0)$. The proof does not use $\text{fst}(\psi(x)) = x$: the combinations $F_1 \dots F_n$ are quantified <i>before</i> the algebra, so one instance with $o_i = o_j$ forces $F_i = F_j$ and another with $o_i \neq o_j$ forces $F_i \neq F_j$. It grants every printed hypothesis and asks the <i>weakest</i> conclusion — an injective homomorphism, with <code>isSubalgebraOf_range</code> proving the image is a subalgebra — so stronger readings of “isomorphic” are refuted a fortiori. The argument previously on record at <code>Combinator.lean:60–72</code> is invalid, as r0044 said: <code>isAlgEmbedding_const_of_subsingleton</code> shows two one-point algebras do land on the same subalgebra. The conclusion was right for a reason nobody had given
27	Theorem 27	P thm27
28	Lemma 28	R refuted at generic U (not_forall_lemma28, witness <code>Flat Empty</code>), and stays false after adding <code>[Domain U]</code> and <code>[BoundedComplete U]</code> . Proved at U : <code>lemma28AtU</code> , all nine conjuncts. The blocker is <code>UniversalForBCD U</code>
29	Theorem 29	p sentence 1 proved (thm29). Sentence 2: <code>Thm29Second</code> refuted — our transcription dropped the paper’s word “domain”. <code>Thm29SecondAtDomains</code> , the true reading, is open . r0049 closed the finite half of what it reduces to: $\aleph$ is universal for the finite pointed posets (R49.Agent5.exists_normal_embedding_Ainf, new and unconditional), giving <code>thm29Normal_finiteBasis</code> . <code>Thm29Normal</code> is now open only for infinite K(E)
30	Lemma 30	R universal closure refuted (not_forall_lemma30). <code>Lemma30AtV</code> is open, now at arity 1 — r0049 discharged <code>FpImagesBifinite V</code> outright (R49.Agent6.fpImagesBifinite_V, no hypotheses, no binders), and conjuncts 1–2 now follow from <code>Thm29SecondAtDomains</code> alone. <code>Thm29Normal</code> is the single remaining named obstruction

Result 15 was never missing — the instruments were. Both the declaration scan and the docstring grep searched only for `thm|theorem|lem|lemma`, and Gunter & Scott number **Propositions in the same sequence**. `scripts/numbered-status.sh` now matches `prop|proposition|cor|corollary` as well; the widening was measured before it was made, and over results 1–30 it adds exactly one hit and no false positive.

Result 2 was genuinely absent, and the two gaps had independent causes: 15 was a grep blind spot, 2 was a real hole. Widening the heading-word set fixes the 15-class defect but **would still have missed 2**.

Results 4, 5 and 8 carry no numbered declaration either, but are each quoted in a module docstring — `NormalSubposet.lean`, `FinitaryProjection.lean`, `Product.lean` — so **a result stated under a descriptive name is the normal case here**, and a missing number is not evidence of a missing proof.

The figure “30 numbered results” is now verified, by extracting every `(Theorem|Lemma|Proposition|Corollary) N` heading from the full text — ASCII headings are unaffected by the Type 3 font defect. Exactly 30, contiguous, no gap and no repeat. The single Proposition among 29 is the entire cause of this round’s measurement defect.

What is open, in one table

#	Item	What it needs
1	<code>ScottHomCRecursive</code>	RecursiveNormal for $K(D \rightarrow E)$ — the one item r0049 did not close. Deciding whether two basis elements of $D \rightarrow E$ are bounded, and where their join is. <code>StepFunctionsDecidable</code> is restated and discharged from it (R49.Agent3.stepFunctionsDecidable_of_scottHomC); the order test needs no search at all (R49.Agent4.computablePred_le_stepValues)

#	Item	What it needs
2	Theorem7ArrowRecursive	item 1
3	Theorem7StrictRecursive	item 1
4	Thm29Normal at infinite K(E)	the copies must nest under incl, so the type must be realized over the <i>tower's</i> image, not the η -image. Finite bases are closed
5	Thm29SecondAtDomains	item 4
6	Lemma30AtV	item 4, and nothing else — FpImagesBifinite V is discharged
7	Lem30Arrow	item 6
8	PreservesRecursivePresentation	proved at fstOp/sndOp; open at the arrow, where it is equivalent to item 2
9	the eight open $S \neq$ rows	five of them (Universality.lem24, Universality.thm25) are one obstruction, not five: whether the closure image is algebraic — a closure image on an algebraic lattice is continuous, not algebraic

Items 2–3 and 5–7 are downstream of 1 and 4. Only items 1, 4 and 9 are real work. Item 1 is no longer mechanical and item 4 is no longer the whole of Thm29Normal.

Additional theories

Results from other authors, formalized here because the paper's proofs need them. None is part of Gunter & Scott's 30.

Jung, *Cartesian Closed Categories of Domains*

#	Result	Status
1	Corollary 1.36	proved — JungCor136, not by Jung's route (which goes through Prop. 1.22 and a retraction pair) but by indexing below cap e
2	Theorem 1.37	proved at [Domain D] — R45.Agent5.thm137. Full generality open
3	Theorem 1.37 for chains	proved at [Domain D] — thm137Chains
4	Proposition 1.22	not needed — the route around it is item 1
5	Theorem 2.1, 2.3	quoted; five printed defects in Jung's write-up recorded

Iwamura and Markowsky

#	Result	Status
1	Iwamura's lemma	proved — exists_chain_directed_cover. Never in Mathlib
2	Markowsky's theorem	proved — hasChainSuprema_iff_hasDirectedSuprema. Never in Mathlib
3	HasWellOrderedInfima	no producer exists — five occurrences, all consumers. A complete reduction chain with an empty left end

Spreen 2005

#	Result	Status
1	Lemma 5.8	proved — PropertyM.hasOmegaOpBoundsAbove_pair. This, not Iwamura, is what closed Jung's 1.37 here

Gunter 1987, *Universal Profinite Domains*

#	Result	Status
1	Lemma 24 at M(A)	proved — R47.Agent1.lemma24_MPair. The first proof anywhere: Gunter's printed Lemma 24 produces <i>some</i> A^+ and is not about M(A); the M(A) form is a remark with no theorem and no proof

#	Result	Status
2	HasNormalRealizations at Ainf	refuted — not_hasNormalRealizations_Ainf. The route to Thm29Normal is sound but its hypothesis is unsatisfiable at this tower. r0049 narrowed what this closes: the refutation quantifies <i>universally</i> over finite normal $A \triangleleft A^\infty$, while Thm29Normal quantifies <i>existentially</i> over embeddings, so it closes extension of an embedding already fixed in A^∞ and not construction of one. That is what let lemma24_Step be run inside the stages
3	Proposition 21, Theorems 22 and 25	used, via the implication thm29Normal_of_hasNormalRealizations

Defects in the printed paper

Ten, nine recorded in StatementRecovery.md. The tenth is r0049's: **Theorem 26 is false as printed** for any signature with two or more 0-ary slots, kernel-checked by R49.Agent7.not_thm26Printed_of_two_zero_aritys. That row sat in the suspected-and-refuted table below until r0049 because **the reason on record for it was wrong**, which is a separate defect from the claim being wrong.

Two suspected defects remain **our** transcription errors, not the paper's:

#	Suspected	Actually
1	Theorem 29's second sentence false	our transcription dropped "domain" from "any bifinite domain"
2	the step-function enumeration	our guard tests compactness where the paper tests boundedness — and r0049 sharpened this: on the paper's own index the join always exists, so the paper tests neither

An **eleventh candidate is not yet recorded**: r0049's agent3 reports that the same folio-12 sentence writes $\bigcup\{f(y) \mid y \in N \cap \downarrow x\}$ where f is bound nowhere and s is meant. It is not counted above and not yet in StatementRecovery.md.

Reproducing

```
scripts/counts.sh          # modules, lines, theorems, sorry
scripts/numbered-status.sh # declarations per numbered result
scripts/a6-env-scan.sh <out> # then a6-summarize.py for row 11
scripts/a5-r47-conditional.sh # row 8, the S+H conditional surface
scripts/compile.sh -r <round> # build
```